

$$\begin{aligned}
 P(C|E) &= P(C|J, S) P(J, S|E) + \\
 &\quad P(C|\sim J, S) P(\sim J, S|E) + \\
 &\quad P(C|J, \sim S) P(J, \sim S|E) + \\
 &\quad P(C|\sim J, \sim S) P(\sim J, \sim S|E) \\
 &= 0.95 * P(J|E) * P(S|E) + \\
 &\quad 0.9 * P(\sim J|E) * P(S|E) + \\
 &\quad 0.9 * P(J|E) * P(\sim S|E) + \\
 &\quad 0.1 * P(\sim J|E) * P(\sim S|E)
 \end{aligned}$$

$$\begin{aligned}
 &= 0.95 * 0.8 * 0.1 + \\
 &\quad 0.9 * 0.2 * 0.1 + \\
 &\quad 0.9 * 0.8 * 0.9 + \\
 &\quad 0.1 * 0.2 * 0.9
 \end{aligned}$$

$$= 0.76$$

$$P(E|C) = \frac{P(C|E) * P(E)}{P(C)}$$

$$= \frac{0.76 * 0.5}{P(C|E) * P(E) + P(C|\sim E) * P(\sim E)}$$

$$= \frac{0.38}{0.38 + P(C|\sim E) * 0.5}$$

$$P(C|\sim E) = P(C|J, S) * P(J, S|\sim E) + \\ P(C|\sim J, S) * P(\sim J, S|\sim E) + \\ P(C|\sim J, \sim S) * P(\sim J, \sim S|\sim E) + \\ P(C|\sim J, \sim S) * P(\sim J, \sim S|\sim E)$$

$$= 0.95 * P(J|\sim E) * P(S|\sim E) + \\ 0.9 * P(\sim J|\sim E) * P(S|\sim E) + \\ 0.9 * P(J|\sim E) * P(\sim S|\sim E) + \\ 0.1 * P(\sim J|\sim E) * P(\sim S|\sim E)$$

$$= 0.95 * 0.1 * 0.5 + \\ 0.9 * 0.9 * 0.5 + \\ 0.9 * 0.1 * 0.5 + \\ 0.1 * 0.9 * 0.5$$

$$= 0.5425$$

(2)

$$P(E|C) = \frac{0.38}{0.38 + 0.5425 \times 0.5}$$

$$= \frac{0.38}{0.65125}$$

$$= 0.58$$